ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 4

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE A

MARKS: 50 **TIME: 1 hour**

An end-of-year test on Grade 4 Terms 3 and 4: Energy and Change (energy for life, food chains, energy around us, movement and energy in a system) and Planet Earth and Beyond (the Earth, the Sun, the Moon and rocket systems).

Written by Academic Central for practice, following the CAPS curriculum for Grade 4. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully before you answer.
4. Write neatly. You may use a pencil for the answers that ask you to draw.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.9 B.

1.1 Plants get the energy they need to make their food from …

A the soil. B the Sun. C water. D animals.

1.2 Which of these gives us energy when we eat it?

A Food B Petrol C Wind D Coal

1.3 An animal that eats only plants is called a …

A carnivore. B herbivore. C omnivore. D producer.

1.4 A torch changes electrical energy into …

A light energy. B food energy.
C wind energy. D energy for our muscles.

1.5 The Earth spins around once every …

A hour. B 24 hours. C week. D year.

1.6 The Earth moves once around the Sun every …

A day. B month. C 365¼ days. D 28 days.

1.7 The Sun is a …

A planet. B moon. C star. D rock.

1.8 We can see the Moon at night because it …

A makes its own light. B reflects light from the Sun.
C is on fire. D is closer to us than the stars.

[8]

QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number.

2.1 Animals get energy from the food they eat.

2.2 The Moon is bigger than the Earth.

2.3 We should never look straight at the Sun.

2.4 A wound-up elastic band stores energy.

2.5 When it is day in South Africa, it is day everywhere on Earth.

2.6 A rocket pushes hot gases downwards so that it moves upwards.

[6]

QUESTION 3: ENERGY FOR LIFE

Look at the food chain in FIGURE 1 and answer the questions.

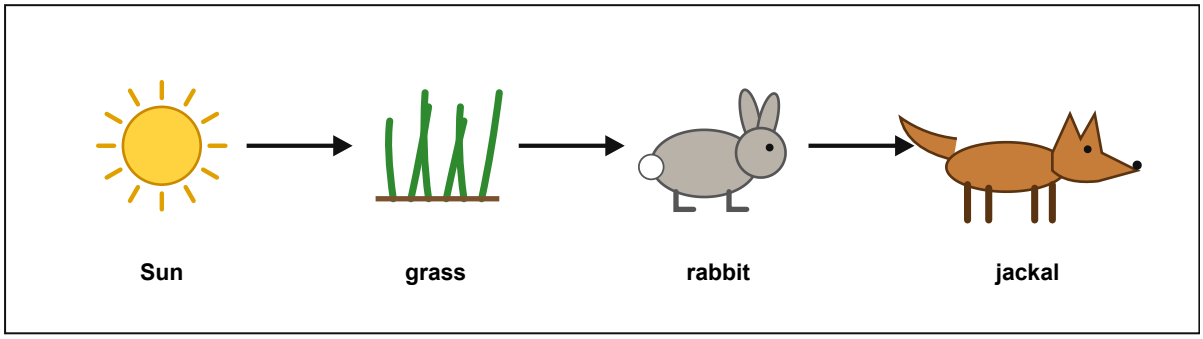


FIGURE 1: A food chain

3.1 Where does all the energy in this food chain come from first? (1)

3.2 Name the producer in this food chain. (1)

3.3 Why do we call grass a producer? (1)

3.4 Name the herbivore in this food chain. (1)

3.5 Name the carnivore in this food chain. (1)

3.6 What do the arrows in a food chain show? (1)

3.7 What would happen to the jackals if all the rabbits died? Give a reason for your answer. (2)

3.8 People are omnivores. Explain what this means, and give ONE example of a meal that shows it. (2)

3.9 Give TWO reasons why our bodies need energy from food. (2)

[12]

QUESTION 4: ENERGY AROUND US AND MOVEMENT

4.1 Besides food, name TWO sources of energy that make things work in a home. (2)

4.2 Complete the energy chain for a torch: stored energy in the battery → _____ energy → light energy (1)

4.3 Thandi made the toy car in FIGURE 2. It moves when the elastic band unwinds.

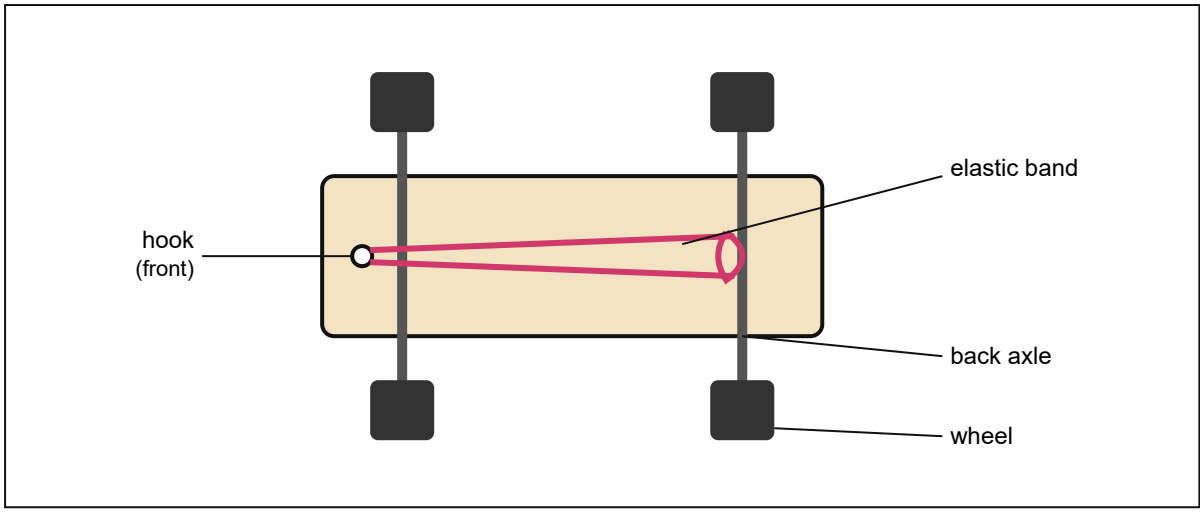


FIGURE 2: Thandi's toy car, seen from above

4.3.1 What must Thandi do to the elastic band to give the car energy? (1)

4.3.2 What is the input (the energy put in) of this system? (1)

4.3.3 What is the output of this system? (1)

4.3.4 Give ONE way Thandi could make her car travel further. (1)

4.4 Explain why you should never push your fingers or any object into a plug socket. (2)

4.5 Give ONE way to save electricity at home. (1)

[10]

QUESTION 5: PLANET EARTH, THE SUN AND THE MOON

FIGURE 3 shows the Sun shining on the Earth. A and B are two children in different places on Earth.

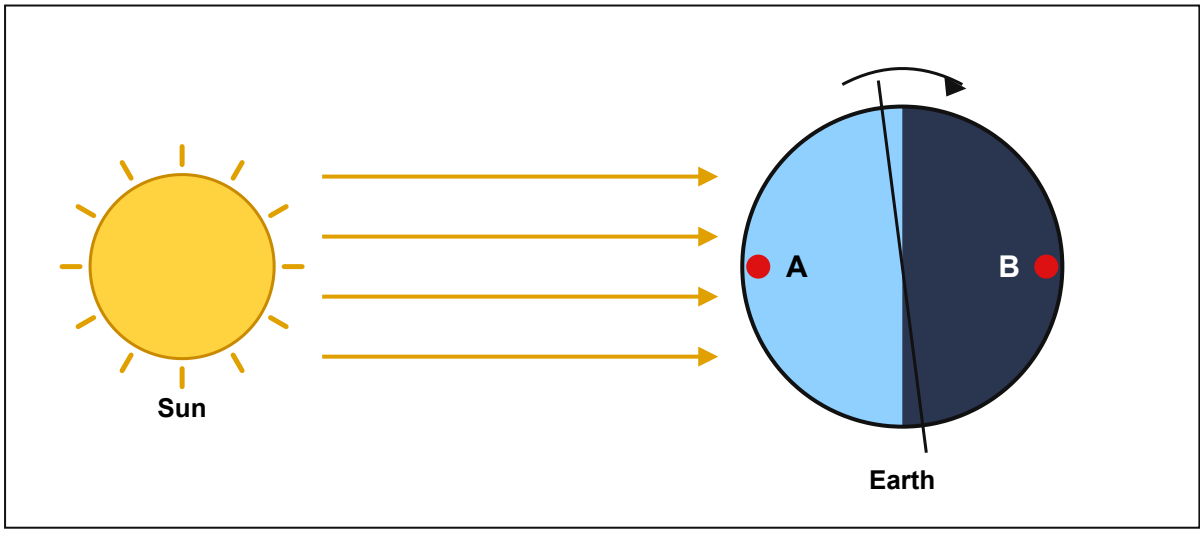


FIGURE 3: The Sun and the Earth

5.1 Is it day or night for child A? (1)

5.2 Is it day or night for child B? (1)

5.3 Explain why we have day and night. (2)

5.4 What is the shape of the Earth? (1)

5.5 Give TWO ways the Sun is important for life on Earth. (2)

5.6 Why must you never look straight at the Sun? (1)

5.7 The Moon moves around the Earth. About how long does it take to go around the Earth once? (1)

5.8 The shape of the Moon that we see seems to change during the month. What do we call these different shapes? (1)

5.9 Give TWO differences between the Earth and the Moon. (2)

5.10 Astronauts travel to space in rockets.

5.10.1 Why must astronauts wear spacesuits on the Moon? (1)

5.10.2 What makes a rocket move upwards? (1)

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TOTAL: 50

END OF QUESTION PAPER